

$$= 126800 \left( 1 + \frac{3}{20} \right) \left( 1 - \frac{1}{5} \right)$$

$$= 126800 \left( \frac{23}{20} \right) \left( \frac{4}{5} \right) = 116656$$

35. (c) Given,  $R_1 = 15\%$ ,  $R_2 = 15\%$

$$\therefore \text{Population} = P \left( 1 + \frac{R_1}{100} \right) \left( 1 - \frac{R_2}{100} \right)$$

$$= 126800 \left( 1 + \frac{15}{100} \right) \left( 1 - \frac{15}{100} \right)$$

$$= 126800 \times \frac{115}{100} \times \frac{85}{100} = 123947$$

36. (b) Given,  $R_1 = 15\%$  and  $R_2 = 20\%$

$\therefore$  Required population

$$= P \left( 1 + \frac{15}{100} \right) \left( 1 + \frac{20}{100} \right)$$

$$= 126800 \left( \frac{23}{20} \right) \left( \frac{6}{5} \right) = 174984$$

37. (c) Refer to question 21.

38. (b) Refer to question 34.

39. (a) 5% of 50% of 500
- $$= \frac{5}{100} \times \frac{50}{100} \times 500 = 12.5 \text{ [Rule 1]}$$

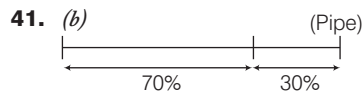
40. (d) Given, the cost of meal of Y = ₹ 100  
Now, according to the question,  
Cost of the meal of Z = 20% more than that of Y
- $$= \left( 100 + \frac{20}{100} \times 100 \right)$$

$$= (100 + 20) = ₹ 120$$

and, the cost of the meal of X =  $\frac{5}{6}$  as much as the cost of the meal of Z =  $\frac{5}{6} \times 120 = ₹ 100$

$\therefore$  Total amount that all the three of them has paid

$$= 100 + 100 + 120 = ₹ 320$$



$\therefore$  Increase in percentage of longer piece compared to shorter piece

$$= \frac{70 - 30}{30} \times 100\% \quad [\text{Rule 4}]$$

$$= \frac{40}{30} \times 100\% = \frac{400}{3}\%$$

42. (c) Refer to question 6.

43. (d) Average price increase
- $$= \left( \frac{5 + 8 + 77}{3} \right) \% = \frac{90}{3} \% = 30\%$$

44. (a) Refer to question 23.

45. (b) Refer to question 2.

46. (d) Refer to question 18.

47. (b) Let the number of voters on the voter list be  $x$ .

$$\text{Total cast vote} = 90\% \text{ of } x - 60$$

Winner was supported by 47% of total voter. i.e. 47% of  $x$ .

Hence, rival got vote = (90% of  $x - 60$ ) - 47% of  $x$ .

$$= 43\% \text{ of } x - 60$$

It is given that difference between their vote is 308.

Then, 47% of  $x - 43\% \text{ of } x - 60 = 308$

$$\Rightarrow 4\% \text{ of } x = 308 - 60$$

$$\Rightarrow \frac{4}{100} x = 248$$

$$\therefore x = \frac{248 \times 100}{4} = 6200$$

48. (c) Let the original salary be  $x$ .

Then, increased salary

$$= \left( \frac{110}{100} \right) x = ₹ \frac{11x}{10}$$

$\therefore$  He received the same salary even after increment.

Amount of salary he did not receive

$$= \frac{11x}{10} - x = ₹ \frac{x}{10}$$

$\therefore$  Amount of salary in percentage

$$= \left( \frac{\frac{x}{10}}{\frac{11x}{10}} \right) \times 100\%$$

$$= \frac{x}{11x} \times 100\% = \frac{100}{11}\%$$

49. (b) Refer to question 31.

# 09

## SIMPLE INTEREST

*Regularly (1-2) questions have been asked from this chapter. Generally direct formula based questions are asked from this chapter and hence making it easy to score area.*



### INTEREST

When money is borrowed by a person, then customarily the money lender used to charge some extra money in lieu of the money lent by him. This extra money earned by the money lender is called interest.

Some terms related to interest are given below

**Principal (P)** The money which is borrowed from a money lender, is called Principal.

**Amount (A)** The sum of the principal and the interest is called Amount i.e. Amount (A) = Principal (P) + Interest (I).

**Rate (R)** It is the rate at which the interest is charged on Principal.

**Time (T)** The time period, for which the money is lent or deposited or borrowed, is called Time.

### SIMPLE INTEREST

If the interest is calculated on the original Principal for any length of time, then it is called simple interest.

$$\text{Simple Interest (SI)} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100} \quad \text{or} \quad \text{SI} = \frac{P \times R \times T}{100}$$

$$\text{Amount (A)} = \text{Principal (P)} + \text{Interest (I)} \quad \text{and} \quad P = \frac{100 \times A}{100 + RT}$$

Where,  $A \rightarrow$  Amount

$R \rightarrow$  Rate of simple Interest

$T \rightarrow$  Time

**EXAMPLE 1.** The sum required to earn a monthly interest of ₹400 at 10% per annum at simple interest is

a. ₹ 2000

b. ₹ 12000

c. ₹ 24000

d. ₹ 48000

**Sol. d.** Given,  $SI = ₹ 400$ ,  $R = 10\%$  and  $T = 1 \text{ month} = \frac{1}{12} \text{ yr}$

$$SI = \frac{P \times R \times T}{100} \quad \text{or} \quad P = \frac{100 \times SI}{R \times T} = \frac{100 \times 400 \times 12}{10 \times 1} = ₹ 48000$$

**EXAMPLE 2.** In what time will the simple interest on ₹ 400 at 10% per annum be the same as the simple interest on ₹ 1000 for 4 yr at 4% per annum?

- a. 2 yr      b. 3 yr      c. 4 yr      d. 6 yr

**Sol. c.** Here,  $P = ₹ 1000$ ,  $T = 4 \text{ yr}$ , and  $R = 4\%$

$$\therefore \text{Simple interest} = \frac{P \times R \times T}{100} = \frac{1000 \times 4 \times 4}{100} = ₹ 160$$

Now, simple interest = ₹ 160,  $P = ₹ 400$ ,  $R = 10\%$ ,

$$\text{then, } T = \frac{100 \times SI}{P \times R} = \frac{100 \times 160}{400 \times 10} = 4 \text{ yr}$$

**EXAMPLE 3.** A sum at simple interest of 4% per annum amounts to ₹ 3120 in 5 yr. Find the sum.

- a. ₹ 2500      b. ₹ 1300      c. ₹ 4000      d. ₹ 2600

**Sol. d.** Here,  $T = 5 \text{ yr}$ ,  $R = 4\%$ ,  $A = ₹ 3120$

$$\text{We know that, } P = \frac{100 \times A}{100 + RT} = \frac{100 \times 3120}{100 + 4 \times 5} = \frac{100 \times 3120}{120} = ₹ 2600$$

## Important Rules and Formulae

**(Rule 1)** If a sum of money becomes  $n$  times in ' $T$ ' yr at simple interest, then rate of interest will be,  
 $R = \left( \frac{100(n-1)}{T} \right) \%$

**EXAMPLE 4.** At what rate per cent per annum will a sum of money double in 8 yr?

- a. 12%      b.  $12\frac{1}{2}\%$   
 c. 13%      d. 15%

**Sol. b.** Here,  $n = 2$  and  $T = 8 \text{ yr}$

$$\therefore \text{Rate} = \frac{100(n-1)}{T} = \frac{100 \times (2-1)}{8} = \frac{100}{8} = 12\frac{1}{2}\%$$

**(Rule 2)** If a certain Principal Amounts to ₹  $A_1$  in  $t_1$  yr and to ₹  $A_2$  in  $t_2$  yr, then the sum ( $P$ ) is given by  
 $₹ \left( \frac{A_2 t_1 - A_1 t_2}{t_1 - t_2} \right)$  and the rate per cent ( $R$ ) per annum is  
 given by  $\left[ \frac{100 (A_2 - A_1)}{A_1 t_2 - A_2 t_1} \right] \%$ .

**EXAMPLE 5.** A certain sum amounts to ₹ 1586 in 2 yr and ₹ 1729 in 3 yr. Find the rate and the sum.

- a. 8%, ₹ 1200  
 b. 9%, ₹ 1300  
 c. 10%, ₹ 1000  
 d. 11%, ₹ 1300

**Sol. d.** Here,  $A_1 = ₹ 1586$ ,  $t_1 = 2 \text{ yr}$ ,  $A_2 = 1729$  and  $t_2 = 3 \text{ yr}$

$$\therefore \text{Required principle} = \frac{A_2 t_1 - A_1 t_2}{t_1 - t_2} = \frac{1729 \times 2 - 1586 \times 3}{2 - 3} = ₹ 1300$$

$$\text{and required rate } R = \left[ \frac{(A_2 - A_1) \times 100}{A_1 t_2 - A_2 t_1} \right] \%$$

$$= \frac{(1729 - 1586) \times 100}{1586 \times 3 - 1729 \times 2} = \frac{143 \times 100}{1300} = 11\%$$

**(Rule 3)** At the same rate of simple interest, if a sum of money becomes  $n_1$  times of itself in  $t_1$  yr and  $n_2$  times in  $t_2$  years, then  $t_2 = \frac{(n_2 - 1)}{(n_1 - 1)} t_1 \text{ yr}$ .

**EXAMPLE 6.** A sum of money becomes 3 times in 5 yr at simple interest. In how many years, will the same sum become 6 times at the same rate of simple interest?

- a. 10 yr  
 b. 12 yr  
 c. 12.5 yr  
 d. 10.5 yr

**Sol. c.** Here,  $n_1 = 3$ ,  $t_1 = 5 \text{ yr}$  and  $n_2 = 6$ ,  $t_2 = ?$

$$\therefore \text{Required time } (t_2) = \frac{(n_2 - 1)}{(n_1 - 1)} t_1 = \frac{(6 - 1)}{(3 - 1)} \times 5 = \frac{25}{2} = 12.5 \text{ yr}$$

## > PRACTICE EXERCISE

- Find the amount on a sum of ₹ 400 for 3 yr at simple interest at 5% per annum.  
 (a) ₹ 460      (b) ₹ 415      (c) ₹ 435      (d) ₹ 412
- Find what sum of money will amount to ₹ 900 in 4 yr at 5% per annum on simple interest?  
 (a) ₹ 750      (b) ₹ 650  
 (c) ₹ 500      (d) ₹ 550
- If a certain sum is doubled in 8 yr on simple interest, in how many years will it be four times?  
 (a) 24 yr      (b) 16 yr      (c) 32 yr      (d) 12 yr
- A sum of money at simple interest amount to ₹ 1260 in 2 yr and ₹ 1350 in 5 yr, then the rate per cent per annum is  
 (a) 30%      (b) 10%      (c) 2.5%      (d) 5%

5. The difference of 13% per annum and 12% of a sum in 1 yr is ₹ 110. Then, the sum is  
(a) ₹ 12000 (b) ₹ 13000 (c) ₹ 11000 (d) ₹ 16000
6. The simple interest on a sum of money at 10% per annum for 6 yr is half the sum. Then, the sum is  
(a) ₹ 5000 (b) not possible (c) ₹ 4000 (d) ₹ 6000
7. The sum which amounts to ₹ 840 in 5 yr at the rate of 8% per annum simple interest is  
(a) ₹  $\frac{100 \times 8 \times 5 + 100}{840}$  (b) ₹  $\frac{100 + 840}{100 + 5 \times 8}$   
(c) ₹  $\frac{100 \times 840}{100 + 5 \times 8}$  (d) ₹  $\frac{840 \times 5 \times 8}{100}$
8. A certain sum at simple interest amounts to ₹ 1040 in 3 yr and to ₹ 1360 in 7 yr. Then, the sum is  
(a) ₹ 750 (b) ₹ 800 (c) ₹ 900 (d) ₹ 1000
9. A sum becomes 6 fold at 5% per annum. At what rate, the sum becomes 12 fold?  
(a) 10% (b) 12% (c) 9% (d) 11%
10. A man borrowed ₹ 40000 at 8% simple interest per year. At the end of second year he paid back certain amount and at the end of fifth year he paid back ₹ 35960 and cleared the debt. What is the amount did he pay back after the second year?  
(a) ₹ 16200 (b) ₹ 17400  
(c) ₹ 18600 (d) None of these
11. A sum of ₹ 1550 was lent partly at 5% and partly at 8% simple interest. The total interest received after 3 yr was ₹ 300. The ratio of money lent at 5% to 8% is  
(a) 11 : 12 (b) 16 : 15 (c) 12 : 21 (d) 11 : 13
12. Rahim buys a house and pays ₹ 8000 cash and ₹ 9600 at 5 yr credit at 4% per annum simple interest. Then, the cash price of the house  
(a) ₹ 10000 (b) ₹ 9600 (c) ₹ 17000 (d) ₹ 16000
13. At what rate per cent per annum simple interest, will a sum of money triple itself in 25 yr?  
(a) 8% (b) 9% (c) 10% (d) 12%
14. A man invested ₹ 1000 on a simple interest at a certain rate and ₹ 1500 at 2% higher rate. The total interest in 3 yr is ₹ 390. What is the rate of interest for ₹ 1000?  
(a) 4% (b) 5% (c) 6% (d) 8%
15. In what time the simple interest on a sum of money be  $\frac{3}{8}$  of the principal with rate of interest  $3\frac{1}{8}\%$ ?  
(a) 9 yr (b) 6 yr (c) 12 yr (d) 15 yr
16. If the rate of simple interest is 12% per annum the amount that would fetch interest of ₹ 6000 per annum is  
(a) ₹ 7200 (b) ₹ 72000 (c) ₹ 50000 (d) ₹ 48543.69
17. A lends a sum of money for 10 yr at 5% simple interest, B lends double that amount for 5 yr at the same rate of interest. Which of the following statement is true in this regard?  
(a) A will get twice the amount of interest that B would get  
(b) B will get twice the amount of interest that A would get.  
(c) A and B will get the same amount as interest.  
(d) B will get four times the amount of interest that A would get
18. Out of a sum of ₹ 625 a part was lent at 5% and the other at 10% simple interest. If the interest on the first part after 2 yr is equal to the interest on the second part after 4 yr, then the second sum (in ₹) is  
(a) ₹ 125 (b) ₹ 200 (c) ₹ 250 (d) ₹ 300
19. A man invests an amount of ₹ 15860 in the names of his three sons A, B and C in such a way that they get the same interest after 2, 3 and 4 yr, respectively. If the rate of simple interest is 5%, then the ratio of the amounts invested among A, B and C will be  
(a) 10 : 15 : 20 (b)  $\frac{1}{10} : \frac{1}{15} : \frac{1}{20}$   
(c) 110 : 115 : 120 (d)  $\frac{1}{110} : \frac{1}{115} : \frac{1}{120}$
20. A sum was put at simple interest at a certain rate for 2 yr. Had it been put at 3% higher rate, it would have fetched ₹ 72 more. The sum is  
(a) ₹ 1200 (b) ₹ 1600 (c) ₹ 1900 (d) ₹ 1400
21. Harsha makes a fixed deposit of ₹ 20000 in Bank of India for a period of 3 yr. If the rate of interest be 13% SI per annum charged half yearly, what amount will he get after 42 months?  
(a) ₹ 27800 (b) ₹ 28100 (c) ₹ 29100 (d) ₹ 30000
22. A sum of money lent on simple interest triples itself in 15 yr and 6 months. In how many year still it be doubled?  
(a) 6 yr and 3 months (b) 7 yr and 9 months  
(c) 8 yr and 3 months (d) 9 yr and 6 months
23. Mr Pawan invests an amount of ₹ 24200 at the rate of 4% per annum for 6 yr to obtain a simple interest, later he invests the principal amount as well as the amount obtained as simple interest for another 4 yr at the same rate of interest. What amount of simple interest will be obtained at the end of the last 4 yr?  
(a) ₹ 4800 (b) ₹ 4850.32 (c) ₹ 4801.28 (d) ₹ 4700
24. A person invested some amount at the rate of 12% simple interest and the remaining at 10%. He received yearly an interest of ₹ 130. Had he interchanged the amounts invested, he would have received an interest of ₹ 134. How much money did he invest at different rates?  
(a) ₹ 500 at the rate of 10%, ₹ 800 at the rate of 12%  
(b) ₹ 700 at the rate of 10%, ₹ 600 at the rate of 12%  
(c) ₹ 800 at the rate of 10%, ₹ 400 at the rate of 12%  
(d) ₹ 700 at the rate of 10%, ₹ 500 at the rate of 12%

**Directions** (Q. Nos. 25-27) If SI for a certain sum  $P_1$  for time  $T_1$  and rate of interest  $R_1$  is  $I_1$  and SI for another sum  $P_2$  for time  $T_2$  and rate of interest  $R_2$  is  $I_2$ , then difference of SI =  $I_2 - I_1 = \frac{P_2 R_2 T_2 - P_1 R_1 T_1}{100}$ .

- 25.** Simple interest for the sum of ₹ 1500 is ₹ 50 in 4 yr and ₹ 80 in 8 yr, the rate of SI is  
(a) 0.6% (b) 5% (c) 0.05% (d) 0.5%
- 26.** Simple interest for the sum of ₹ 1230 for 2 yr is ₹ 10 more than the simple interest for ₹ 1130 for the same duration. Find the rate of interest.  
(a) 5% (b) 6% (c) 8% (d) 2%
- 27.** If the annual payment of ₹  $A$  will discharge a debt of ₹ 1092 due in 2 yr at 12% simple interest, then  
I.  $A$  will be ₹ 515 (approx.).  
II.  $A$  will be ₹ 530 (approx.), if interest rate is 6%.  
(a) Only I (b) Both I and II  
(c) Only II (d) Neither I nor II

## PREVIOUS YEARS' QUESTIONS

- 28.** The principal on which a simple interest of ₹ 55 will be obtained after 9 months at the rate of  $3\frac{2}{3}\%$  per annum is **2013 I**  
(a) ₹ 1000 (b) ₹ 1500  
(c) ₹ 2000 (d) ₹ 2500
- 29.** In how much time would the simple interest on a principal amount be 0.125 times the principal amount at 10% per annum? **2015 I**  
(a)  $1\frac{1}{4}$  yr (b)  $1\frac{3}{4}$  yr (c)  $2\frac{1}{4}$  yr (d)  $2\frac{3}{4}$  yr
- 30.** If a sum of money at a certain rate of simple interest per year doubles in 5 yr and at a different rate of simple interest per year becomes three times in 12 yr, then the difference in the two rates of simple interest per year is **2016 I**  
(a) 2% (b) 3% (c)  $3\frac{1}{3}\%$  (d)  $4\frac{1}{3}\%$

## ANSWERS

|    |   |    |   |    |   |    |   |    |   |    |   |    |   |    |   |    |   |    |   |
|----|---|----|---|----|---|----|---|----|---|----|---|----|---|----|---|----|---|----|---|
| 1  | a | 2  | a | 3  | a | 4  | c | 5  | c | 6  | b | 7  | c | 8  | b | 9  | d | 10 | b |
| 11 | b | 12 | d | 13 | a | 14 | a | 15 | c | 16 | c | 17 | c | 18 | a | 19 | b | 20 | a |
| 21 | c | 22 | b | 23 | c | 24 | d | 25 | d | 26 | a | 27 | b | 28 | c | 29 | a | 30 | c |

## HINTS AND SOLUTIONS

- 1.** (a) Given,  $P = ₹ 400$ ,  $R = 5\%$  and  $T = 3$  yr

$$\text{Simple interest} = \frac{P \times R \times T}{100}$$

$$SI = \frac{400 \times 5 \times 3}{100} = ₹ 60$$

$$\therefore \text{Amount} = P + SI = 400 + 60 = ₹ 460$$

- 2.** (a) Let the sum of money be ₹  $x$ .

$$\therefore \text{Amount} = \text{Sum} + SI$$

$$\therefore \text{Amount} = x + \frac{x \times 5 \times 4}{100}$$

$$\left[ \because SI = \frac{P \times R \times T}{100} \right]$$

$$\text{But amount} = ₹ 900$$

$$\therefore 900 = x + \frac{20x}{100}$$

$$\Rightarrow 900 = \frac{6x}{5} \Rightarrow x = \frac{900 \times 5}{6} = ₹ 750$$

**Shortcut Method**

$$\text{Here, } P = ₹ 900, R = 5\% \text{ and } T = 4 \text{ yr}$$

$\therefore$  Sum

$$= \frac{100 \times A}{100 + RT} = \frac{100 \times 900}{100 + 5 \times 4} = ₹ 750$$

- 3.** (a) Let the sum be ₹  $x$ , so amount = ₹  $2x$

$$\therefore SI = ₹ x$$

Let  $R$  be rate of interest.

$$\therefore R = \frac{100 \times SI}{P \times T} = \frac{100 \times x}{x \times 8} = 12.5\%$$

Now, the needed amount = ₹  $4x$

$$\therefore SI = ₹ (4x - x) = ₹ 3x$$

$$\therefore T = \frac{100 \times SI}{P \times R} = \frac{100 \times 3x}{x \times 12.5} = 24 \text{ yr}$$

**Shortcut Method**

$$\text{Here, } n_1 = 2, n_2 = 4 \text{ and } t_1 = 8 \text{ yr}$$

$$\therefore \text{Required time} = \frac{(n_2 - 1)}{(n_1 - 1)} \times t_1$$

$$= \frac{(4 - 1)}{(2 - 1)} \times 8 = \frac{3}{1} \times 8 = 24 \text{ yr} \quad [\text{Rule 3}]$$

- 4.** (c) Simple interest in 3 yr

$$= ₹ (1350 - 1260) = ₹ 90$$

$\therefore$  Simple interest for 2 yr

$$= \frac{2}{3} \times 90 = ₹ 60$$

$\therefore$  Principal = ₹  $(1260 - 60) = ₹ 1200$

$\therefore$  Rate,

$$R = \frac{100 \times SI}{P \times T} = \frac{100 \times 60}{1200 \times 2} = \frac{60}{24} = \frac{5}{2}$$

$$R = 25\%$$

**Shortcut Method**

$$\text{Here, } A_1 = ₹ 1260, t_1 = 2 \text{ yr,}$$

$$A_2 = ₹ 1350 \text{ and } t_2 = 5 \text{ yr}$$

$\therefore$  Required rate

$$= \left[ \frac{(A_2 - A_1)}{A_1 t_2 - A_2 t_1} \right] \times 100\% \quad [\text{Rule 2}]$$

$$= \frac{(1350 - 1260)}{1260 \times 5 - 1350 \times 2} \times 100\% = 25\%$$